Analyzing Cross-Lingual Topic Spaces and Story Representations in Pre-Trained Language Models

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1 Introduction

2 Related work

Conneau et al. (2020) Eldan and Li (2023) Adelani et al. (2022)

3 Methodology

4 Data

5 Feasible tests

6 Results

A Figures and Tables

Refer to Table 1 to 4 for the tables and Figure 1 and 2 for the two figures.

Language	DeBERTa	Qwen	XLM-RoBERTa
Acholi	0.1994	0.0989	0.0438
Luganda	0.1933	0.1253	0.0853
Masaaba	0.1800	0.0817	0.0814
Nyankole	0.1924	0.1210	0.0865
Soga	0.1911	0.1170	0.0946
Akan	0.1891	0.1277	0.0651
Ewe	0.1813	0.0633	0.0527
Dagbani	0.2281	0.0730	0.0509
Norwegian	0.1251	0.0663	0.0744
Slovak	0.1232	0.0656	0.0751
Serbian	0.1313	0.1568	0.1456
Bulgarian	0.1145	0.0710	0.0636
Czech	0.1216	0.0634	0.0741
Turkish	0.1389	0.0700	0.0819
Persian	0.1374	0.0818	0.0657
Vietnamese	0.1361	0.0614	0.0720
Indonesian	0.0958	0.0566	0.0727
Assamese	0.1815	0.0419	0.0648
Dogri	0.1576	0.0957	0.0506
GOM	0.1757	0.0730	0.0554
Kannada	0.1461	0.0429	0.1174
Nepali	0.1900	0.0551	0.0296
Tamil	0.1330	0.0468	0.0689
Sindhi	0.1466	0.0534	0.0461

Table 1: Silhouette Scores per Language Across Models

Language 1	Language 2	Mean Cosine Sim.
Acholi	Luganda	0.9508
Acholi	Masaaba	0.9619
Acholi	Nyankole	0.9583
Acholi	Soga	0.9615
Acholi	Akan	0.9490
Acholi	Ewe	0.9513
Acholi	Dagbani	0.9727
Acholi	Norwegian	0.6627
Acholi	Slovak	0.6109
Acholi	Serbian	0.6337
Acholi	Bulgarian	0.5994
Acholi	Czech	0.5950
Acholi	Turkish	0.6592
Acholi	Persian	0.7178
Acholi	Vietnamese	0.4640
Acholi	Indonesian	0.6435
Acholi	Assamese	0.7712
Acholi	Dogri	0.8890
Acholi	GOM	0.9279
Acholi	Kannada	0.7280
Acholi	Nepali	0.7242
Acholi	Tamil	0.7609
Acholi	Sindhi	0.7811
Luganda	Nyankole	0.9677
Luganda	Soga	0.9679
Norwegian	Slovak	0.9842
Norwegian	Serbian	0.9835
Slovak	Czech	0.9935
Kannada	Nepali	0.9748
Nepali	Sindhi	0.9694

Table 2: Cross-Lingual Cosine Similarity Metrics (DeBERTa Model)

Language 1	Language 2	Mean Cosine Sim.
Acholi	Luganda	0.9033
Acholi	Masaaba	0.8999
Acholi	Nyankole	0.9018
Acholi	Soga	0.9156
Acholi	Akan	0.8560
Acholi	Norwegian	0.5413
Acholi	Bulgarian	-0.1191
Acholi	Persian	-0.1207
Acholi	Kannada	-0.0689
Luganda	Nyankole	0.9606
Norwegian	Slovak	0.4964
Slovak	Czech	0.8998
Dogri	GOM	0.9543
Kannada	Tamil	0.5917

Table 3: Cross-Lingual Cosine Similarity Metrics (Qwen Model)

References

David Ifeoluwa Adelani, Jade Abbott, Graham Neubig, Ignatius Ezeani Duruaku, and 1 others. 2022.

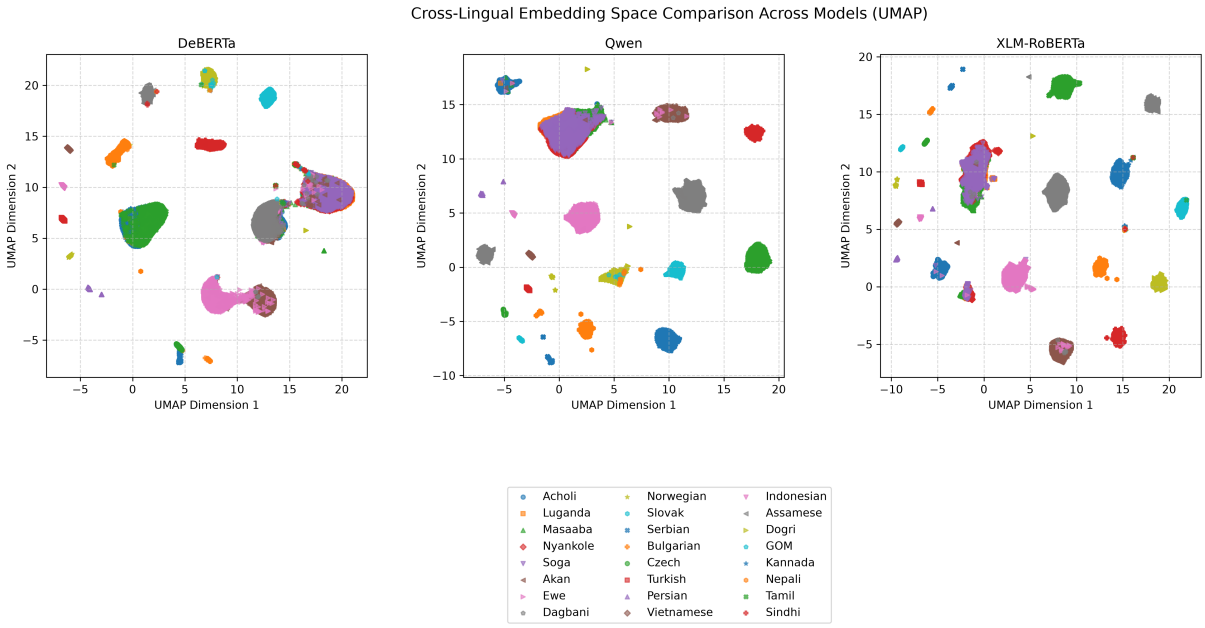


Figure 1: A minimal working example to demonstrate how to place two images side-by-side.

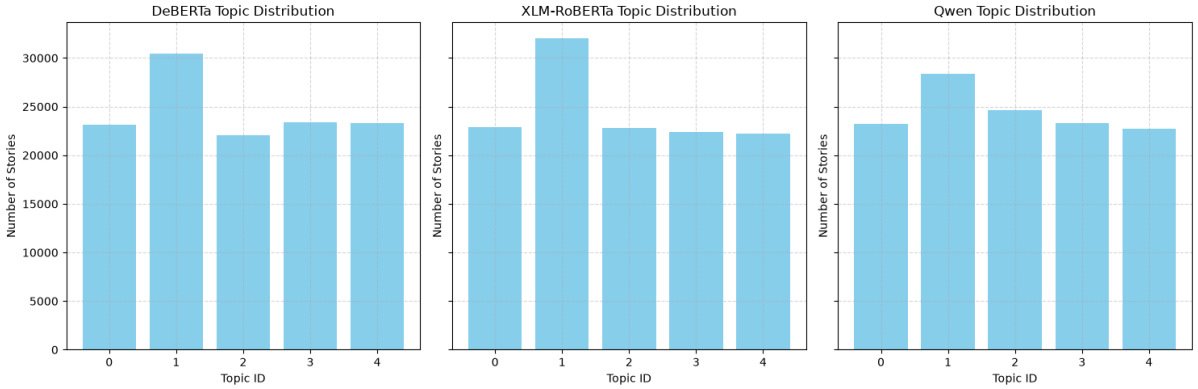


Figure 2: LDA topic categorization distribution

Language 1	Language 2	Mean Cosine Sim.
Acholi	Luganda	0.9965
Acholi	Masaaba	0.9962
Acholi	Nyankole	0.9967
Acholi	Norwegian	0.9894
Luganda	Nyankole	0.9971
Norwegian	Slovak	0.9974
Slovak	Czech	0.9987
Vietnamese	Indonesian	0.9983
Dogri	GOM	0.9980
Kannada	Tamil	0.9975

Table 4: Cross-Lingual Cosine Similarity Metrics (XLM-RoBERTa Model)

MasakhaNEWS: News topic classification for african languages. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*.

Alexis Conneau, Kartik Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Emerging cross-lingual structure in pretrained language models. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*.

Ronen Eldan and Yuanzhi Li. 2023. Tinystories: How small can language models be and still speak coherent english? *arXiv preprint arXiv:2305.07759*.

	Language	Comparison	Dimension 1	Dimension 2	Dimension 3
0	Acholi	DeBERTa vs Qwen	0.999996	0.994984	0.996444
1	Acholi	DeBERTa vs XLM-RoBERTa	0.996180	0.983579	0.979210
2	Luganda	DeBERTa vs Qwen	0.997958	0.996214	0.992004
3	Luganda	DeBERTa vs XLM-RoBERTa	0.992680	0.980998	0.978608
4	Masaaba	DeBERTa vs Qwen	0.996729	0.994966	0.990517
5	Masaaba	DeBERTa vs XLM-RoBERTa	0.993225	0.980985	0.979801
6	Nyankole	DeBERTa vs Qwen	0.994315	0.991878	0.984842
7	Nyankole	DeBERTa vs XLM-RoBERTa	0.988397	0.982236	0.973999
8	Soga	DeBERTa vs Qwen	0.996412	0.994938	0.989129
9	Soga	DeBERTa vs XLM-RoBERTa	0.992065	0.984717	0.972391
10	Akan	DeBERTa vs Qwen	0.988052	0.992737	0.988121
11	Akan	DeBERTa vs XLM-RoBERTa	0.988035	0.983507	0.979119
12	Ewe	DeBERTa vs Qwen	0.992485	0.988232	0.986231
13	Ewe	DeBERTa vs XLM-RoBERTa	0.985299	0.987757	0.976501
14	Dagbani	DeBERTa vs Qwen	0.992607	0.984602	0.984641
15	Dagbani	DeBERTa vs XLM-RoBERTa	0.990721	0.978886	0.982787
16	Norwegian	DeBERTa vs Qwen	1.000000	1.000000	1.000000
17	Norwegian	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
18	Slovak	DeBERTa vs Qwen	1.000000	1.000000	1.000000
19	Slovak	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
20	Serbian	DeBERTa vs Qwen	1.000000	1.000000	1.000000
21	Serbian	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
22	Bulgarian	DeBERTa vs Qwen	1.000000	1.000000	1.000000
23	Bulgarian	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
24	Czech	DeBERTa vs Qwen	1.000000	1.000000	1.000000
25	Czech	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
26	Turkish	DeBERTa vs Qwen	1.000000	1.000000	1.000000
27	Turkish	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
28	Persian	DeBERTa vs Qwen	1.000000	1.000000	1.000000
29	Persian	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
30	Vietnamese	DeBERTa vs Qwen	1.000000	1.000000	1.000000
31	Vietnamese	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
32	Indonesian	DeBERTa vs Qwen	1.000000	1.000000	1.000000
33	Indonesian	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
34	Assamese	DeBERTa vs Qwen	0.998580	0.998028	0.997125
35	Assamese	DeBERTa vs XLM-RoBERTa	0.992181	0.993391	0.987680
36	Dogri	DeBERTa vs Qwen	0.998329	0.998787	0.997214
37	Dogri	DeBERTa vs XLM-RoBERTa	0.992667	0.989423	0.984441
38	GOM	DeBERTa vs Qwen	0.998587	0.998678	0.997664
39	GOM	DeBERTa vs XLM-RoBERTa	0.993198	0.991190	0.983822
40	Kannada	DeBERTa vs Qwen	0.994520	0.976164	0.971726
41	Kannada	DeBERTa vs XLM-RoBERTa	0.992557	0.988733	0.984590
42	Nepali	DeBERTa vs Qwen	0.998997	0.998740	0.998223
43	Nepali	DeBERTa vs XLM-RoBERTa	0.991894	0.992935	0.986940
44	Tamil	DeBERTa vs Qwen	0.984171	0.978970	0.974888
45	Tamil	DeBERTa vs XLM-RoBERTa	0.991453	0.987195	0.984323
46	Sindhi	DeBERTa vs Qwen	0.998605	0.999204	0.998230
47	Sindhi	DeBERTa vs XLM-RoBERTa	0.992767	0.991836	0.985045

Table 5: CCA for each language against each model